MandirSetu: Product Requirements Document (PRD)

Version 1.0 | October 2026 | Status: Draft for review Working product name: MandirSetu (temporary; check name and domain availability) **Prepared for:** Product Owner, Development Team, Temple Committee (pilot village)

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1. Summary in Simple Words (for the Temple Committee and Villagers)

What is this? A simple website that works on any mobile phone. It replaces the paper notebook where the temple writes chanda (donations) and expenses.

What will the committee be able to do?

- Write every chanda and every expense in the phone instead of a notebook.
- See at any time: total collected, total spent, and balance left.
- See festival-wise accounts (for example, how much was collected and spent for one festival).
- Download a report to show the whole village.
- Publish festival dates, puja timings and announcements.

What will villagers be able to do?

- Open the temple page on their phone without any login.
- See the festival calendar for the year, with start and end dates.
- Read announcements from the temple committee.

What is safe?

- Only committee members with a login can enter or change money records.
- Every change is recorded (who changed what, and when). Nothing is silently erased.
- Donor names can be kept private.
- Data is backed up regularly.

What comes later (not in the first version)? QR payment display and online payment, voice assistant in Hindi and Bhojpuri, more temples and villages, SMS or WhatsApp alerts.

First step: One pilot in our village temple. Only after the committee is happy will we offer it to other temples.

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2. Problem Statement

Village temples record chanda and expenses in paper notebooks. This causes:

- **Poor transparency:** donors and villagers cannot easily see how money was used.
- **Errors and disputes:** manual totals, overwriting, lost pages, disagreement about amounts.
- **No history:** hard to compare festival accounts across years.
- **Scattered information:** festival dates, fasting rules and puja timings are shared by word of mouth.
- **Dependence on individuals:** records are hard to hand over when committee members change.

3. Vision and Goals

Vision: A trusted, easy digital platform for every village temple, covering accounts, festival information and, in future, voice-based help in local languages.

Version 1 goals

#	Goal	How we will know
G1	Replace the paper register for chanda and expenses at one pilot temple	Committee uses the app for all new entries for at least 30 days
G2	Make accounts transparent and reportable	Festival-wise report generated and matches the notebook
G3	Give villagers correct festival and temple information	Villagers can find a festival date and announcement without help
G4	Build a foundation that supports many temples later	Two test temples are fully isolated from each other

Non-goals for Version 1: online payment gateway, automatic UPI reconciliation, AI or voice assistant, native Android/iOS apps, tax or complex accounting, nationwide self-service registration, automatic festival-date calculation.

4. Users and Personas

Persona	Description	Main needs
Temple Admin (e.g., committee president/secretary)	Responsible for the temple and its members	Add members, publish events, review reports, final approval
Treasurer / Operator	Records money in and out daily	Fast entry on phone, correct totals, receipts, reports
Villager / Devotee	Browses without login, may have low reading comfort	Festival dates, timings, announcements, Hindi language, large readable text
Platform Owner (product owner)	Operates the system	Create temples, manage access, monitor health, calendar data

Design principle: a committee member who only uses a notebook today should learn the basics in minutes. Large buttons, few fields, Hindi labels, one-hand mobile use.

5. Scope of Version 1

5.1 In scope

Module	Requirement
Temple profile	Name, village, district, state, address, contact, public info
Authentication	Secure login, logout, password recovery
Roles	Temple Admin and Treasurer/Operator
Donation register	Record cash and other donations with optional donor name, date, festival, notes
Expense register	Amount, category, date, description, optional bill photo

Module	Requirement
Dashboard	Total collections, expenses, recorded balance for a selected period
Festival calendar	Verified, year-wise festival dates and details
Temple events	Puja schedules, local events, announcements
Reports	Daily/monthly summaries, festival-wise reports, CSV export, printable summary
Languages	Hindi (primary) and English (optional)
Public temple page	Published events, announcements, temple details
Audit history	Track who created, changed or reversed financial records
Operations	HTTPS, database backups, error logging, tested restore

5.2 Out of scope (later releases)

Payment gateway, UPI auto-confirmation, AI chatbot and voice assistant, SMS/WhatsApp notifications, native mobile apps, complex accounting or tax, self-service multi-temple onboarding, automatic date calculation.

> **Accounting rule:** the app shows the *recorded* balance, not a guarantee of actual cash or bank balance. The pilot starts with a documented opening balance, and the committee reconciles app records with real cash and bank statements.

6. Functional Requirements

Priority: **P0** = must have for pilot, **P1** = should have, **P2** = nice to have.

6.1 Authentication and Access (AUTH)

ID	Requirement	Priority
AUTH-1	Users log in with email or phone-based credential and password	P0
AUTH-2	Passwords stored only as secure hashes	P0
AUTH-3	Password recovery flow	P0
AUTH-4	Roles: Temple Admin, Treasurer/Operator	P0
AUTH-5	Treasurer cannot invite admins or change protected roles	P0
AUTH-6	All permission checks enforced on the server, not only in the UI	P0
AUTH-7	Session expiry and logout	P0

6.2 Temple Workspace (TEMPLE)

ID	Requirement	Priority
TEMPLE-1	Platform owner creates the pilot temple workspace	P0
TEMPLE-2	Admin edits temple profile (name, address, contact, photo)	P0
TEMPLE-3	Admin invites and manages members	P0
TEMPLE-4	Admin sets opening balance with effective date	P0
TEMPLE-5	Every temple-owned record carries a temple ID for future multi-temple use	P0

6.3 Donation (Chanda) Register (DON)

ID	Requirement	Priority
DON-1	Add donation: amount, date/time, payment method (cash, UPI/other), donor name (optional), anonymous flag, festival (optional), notes	P0
DON-2	Auto-generated unique reference number per entry (receipt number)	P0
DON-3	List with search, date filter, festival filter, payment-method filter, pagination	P0
DON-4	View donation details and printable/shareable receipt	P1
DON-5	Correct allowed fields with audit entry; no silent overwrite	P0
DON-6	Reverse an erroneous entry (reversal keeps original visible)	P0
DON-7	Prevent duplicate submission (double tap, retry)	P0
DON-8	Donor identity visible only to authorized roles	P0
DON-9	Payment method and reference fields present now so UPI/QR can be added later without redesign	P0

6.4 Expense Register (EXP)

ID	Requirement	Priority
EXP-1	Add expense: amount, date, category, description, festival (optional), vendor (optional)	P0
EXP-2	Configurable categories (flowers, decoration, food, electricity, priest, etc.)	P0
EXP-3	Attach bill photo/PDF (type and size limited, private storage)	P1
EXP-4	List with filters; correction with audit; reversal	P0

ID	Requirement	Priority
EXP-5	Bills downloadable only by authorized members	P0

6.5 Dashboard and Reports (REP)

ID	Requirement	Priority
REP-1	Dashboard: total collections, total expenses, recorded balance, recent activity for chosen period	P0
REP-2	Balance = opening balance + recorded inflows − recorded outflows	P0
REP-3	Daily, monthly and yearly collection and expense reports	P0
REP-4	Festival-wise report (collections vs expenses)	P0
REP-5	Opening balance, inflows, outflows, closing balance summary	P0
REP-6	CSV export respecting user permissions	P0
REP-7	Printable summary the committee can show villagers, without donor-private details	P1

6.6 Festival Calendar and Events (FEST)

ID	Requirement	Priority
FEST-1	Year-wise festival calendar with name, date, start/end time where applicable, region/tradition, source and verification status	P0
FEST-2	Fasting guidance (when to fast, when to break fast) shown only with a source and tradition	P1
FEST-3	Temple events: title, description, schedule, publish status (draft/published)	P0
FEST-4	Announcements with draft/published status	P0

ID	Requirement	Priority
FEST-5	Link events to general calendar entries (one festival can have many temple events)	P1
FEST-6	Annual calendar update process by the platform owner	P0
FEST-7	Dates are never auto-calculated by the system or AI. They come from a verified source and are checked by the committee/pandit	P0

6.7 Public Pages (PUB)

ID	Requirement	Priority
PUB-1	Public temple page by slug (no login)	P0
PUB-2	Public festival calendar and event details	P0
PUB-3	Only published information is exposed. No donor data, no financial details unless the committee publishes a summary	P0
PUB-4	Hindi/English language switch	P0
PUB-5	Fast loading on low-end phones and slow networks	P0

6.8 Audit and Administration (AUD)

ID	Requirement	Priority
AUD-1	Audit log for every financial create/edit/reverse: who, what, when	P0
AUD-2	Admin can view audit history	P0
AUD-3	Logs must not contain passwords or unnecessary personal data	P0

7. Screen Inventory

Public

ID	Screen	Route	Contents
P01	Home	`/`	Temple search, upcoming festivals, announcements, language selector
P02	Temple profile	`/temples/{slug}`	Details, contact, events, notices
P03	Festival calendar	`/festivals`	Monthly/yearly view, dates, calendar context
P04	Event details	`/events/{id}`	Date, schedule, approved announcements

Private (committee)

ID	Screen	Route	Contents
A01	Login / recovery	`/login`	Secure access
A02	Dashboard	`/admin/dashboard`	Totals, balance, recent activity, quick actions
A03	Donations	`/admin/donations`	List, add, filter, view
A04	Expenses	`/admin/expenses`	List, add, attach bill
A05	Events	`/admin/events`	Create, publish events and announcements
A06	Reports	`/admin/reports`	Summaries, festival-wise, CSV
A07	Settings and members	`/admin/settings`	Profile, opening balance, members, audit history

Mobile navigation: four bottom tabs: **Home, Donations, Expenses, More**. Festivals, reports, members and settings live under More. "Add donation" and "Add expense" are always prominent buttons.

8. Key User Journeys

Journey A: Committee records chanda. Treasurer logs in, taps *Add donation*, enters amount, selects festival, saves. App shows receipt number. Dashboard total updates.

Journey B: Committee records expense. Treasurer taps *Add expense*, selects category, enters amount, optionally photographs the bill, saves.

Journey C: Festival closing. Admin selects festival and period, reviews collections and expenses, resolves discrepancies, exports the report, optionally publishes an approved summary.

Journey D: Villager checks festival. Villager opens the temple link, switches to Hindi, opens the calendar, taps a festival to see dates and timings.

Journey E: Annual update. Platform owner loads next year's verified calendar entries; committee reviews and publishes temple events.

9. Data Model

Use PostgreSQL. Money uses `NUMERIC(12,2)`, never floating-point. Every temple-owned table has `temple_id`.

Entity	Key fields
temples	id, name, slug, village, district, state, address, timezone, default_language, status, created_at
users	id, name, email/phone, password_hash, preferred_language, status, created_at
temple_memberships	id, temple_id, user_id, role, status, created_at
donations	id, temple_id, amount, currency, donated_at, donor_name (nullable), is_anonymous, payment_method, festival_event_id (nullable), reference_number, notes, status (posted/reversed), created_by, updated_at
expenses	id, temple_id, amount, currency, spent_at, category_id, description, vendor, festival_event_id (nullable), bill_file_key (nullable), status, created_by, updated_at

Entity	Key fields
expense_categories	id, temple_id, name, active
opening_balances	id, temple_id, cash_amount, bank_amount, effective_date
festival_calendar_entries	id, calendar_code, region, year, festival_name, festival_date, start_time, end_time, source, verification_status, updated_at
festival_events	id, temple_id, calendar_entry_id (nullable), title, description, start_at, end_at, publication_status, created_by
temple_announcements	id, temple_id, title, body, status, published_at, created_by
file_attachments	id, temple_id, storage_key, mime_type, size, uploaded_by
audit_logs	id, temple_id, actor_user_id, entity_type, entity_id, action, timestamp, change_summary

Design rules

1. Posted financial records are never hard-deleted; use reversal entries.
2. General festival data (`festival_calendar_entries`) is separate from temple-specific events (`festival_events`).
3. Financial writes run in database transactions with unique reference numbers and idempotency protection.
4. Store timestamps in UTC; display in the temple's timezone (India Standard Time).
5. Future `payment_transactions` table will be added when a payment provider is integrated, kept separate from the accounting `donations` record.

10. REST API Plan (v1)

Prefix: `/api/v1` . These are proposed contracts. The development team documents final request/response formats in OpenAPI.

Area	Endpoints
Auth	`POST /auth/login`, `POST /auth/logout`, `POST /auth/forgot-password`, `GET /me`
Temple	`POST /temples` (platform owner), `GET/PATCH /temples/{templeId}`, `GET/POST /temples/{templeId}/members`
Donations	`GET/POST /temples/{templeId}/donations`, `GET/PATCH /temples/{templeId}/donations/{id}`, `POST .../donations/{id}/reverse`
Expenses	`GET/POST /temples/{templeId}/expenses`, `GET/PATCH .../expenses/{id}`, `POST .../expenses/{id}/reverse`, `POST /temples/{templeId}/attachments`
Events	`GET/POST /temples/{templeId}/events`, `PATCH .../events/{id}`, `POST .../events/{id}/publish`
Announcements	`GET/POST /temples/{templeId}/announcements`
Festival calendar	`GET /festivals?year=&region=`, `GET /festivals/{id}`
Reports	`GET /temples/{templeId}/dashboard?from=&to=`, `GET .../reports/collections`, `.../reports/expenses`, `.../reports/financial-summary`, `.../reports/collections.csv`
Settings/Audit	`PATCH /temples/{templeId}/settings`, `GET /temples/{templeId}/audit-logs`
Public	`GET /public/temples/{slug}`, public events, announcements and calendar (published data only)

API rules

- Authorization derives temple scope from the authenticated user's membership. A client-supplied `templeId` is never proof of access.

- Verify that referenced objects (events, categories) belong to the same temple.
- Public endpoints are separate from private endpoints and return only published data.
- List endpoints are paginated; filters are validated server-side.
- Consistent error format and request validation on all endpoints.

11. Non-Functional Requirements

Area	Requirement
Security	HTTPS only; hashed passwords; role and tenant checks on the server; rate limiting on login; secure session or token handling; input validation; private file storage
Privacy	Donor details restricted; no donor data on public pages or in logs; anonymous-donor option
Data integrity	Decimal money type; transactions; no hard deletes of posted records; reversals and audit trail
Performance	Pages usable on low-end Android phones and 3G/4G; main pages load in about 3 seconds on typical mobile networks; list pagination
Usability	Mobile-first; large touch targets; Hindi labels; simple forms; clear error messages
Accessibility	Readable font sizes, good contrast, support for system text scaling; icons paired with labels
Reliability	Daily automated database backups; tested restore procedure; error logging and alerts
Maintainability	Modular monolith, automated tests, OpenAPI docs, CI pipeline, database migrations
Localization	All UI text in translation files (Hindi, English); dates and currency formatted for India (₹)
Connectivity	Version 1 requires internet. Draft-saving and offline entry are a Release 1.x consideration after pilot feedback

12. Technical Architecture

Approach: modular monolith. Simple to build, deploy and maintain by a small team. Services are split only if real scale requires it.

```
Devotees / Committee / Platform Admin (mobile browser)
            |
   Responsive Web Frontend (React + TypeScript)
            |
   Spring Boot REST API (/api/v1)
   Authentication, authorization, validation, tenant isolation
            |
   Business modules: temples, members, donations, expenses,
                     festivals, events, reports, audit
            |
   PostgreSQL  +  Private file storage (bills)
```

Layer	Choice
Frontend	React with TypeScript (or a framework the team can maintain), responsive and mobile-first, i18n library
Backend	Java, Spring Boot, Spring Data JPA, Spring Security
Database	PostgreSQL with migrations (Flyway or Liquibase)
File storage	Private object storage for bills
Testing	JUnit, Mockito, API integration tests, tenant-isolation tests
Deployment	Docker, managed hosting, HTTPS, automated backups, monitoring
Documentation	OpenAPI/Swagger for the API

Future AI layer (Release 2, not built now): speech-to-text, retrieval over verified festival and temple data (RAG), LLM answer generation, text-to-speech, in Hindi and Bhojpuri. Festival dates and timings must come from verified records, never from the model's own calculation.

13. Roles and Responsibilities

Role	Responsibility
Product Owner	Requirements, committee communication, priorities, final acceptance
CTO / Architecture	Technical decisions, security and data design, release gates
Developer(s)	Implementation, tests, documentation, deployment
Temple Committee	Provide current workflow, opening balance, verified festival info; test and give feedback
Pandit / calendar verifier	Confirm local festival dates and timings before publication

14. Milestones and Timeline

Target: **6 to 8 weeks of part-time development**, depending on team availability. Do not cut security and accounting tests to meet the date.

Milestone	Focus	Deliverables	Done when
M1: Foundation (Week 1)	Requirements and setup	Committee interview, notebook workflow documented, repo, Spring Boot project, PostgreSQL, migrations, UI wireframes, translation files	App starts locally and connects to the database
M2: Access (Week 2)	Login, roles, workspace	Authentication, temple and membership entities, role permissions, two test temples	Unauthorized users cannot access another temple's records

Milestone	Focus	Deliverables	Done when
M3: Accounting (Weeks 3–4)	Donations and expenses	Create/list/detail, validation, filters, opening balance, balance calculation, audit, reversals, tests	Committee can enter sample data and totals reconcile
M4: Public site (Week 5)	Temple page and festival info	Public temple page, announcements, year-wise calendar, event management, Hindi/English UI	Villagers can find approved info without login
M5: Reports (Week 6)	Dashboard, exports, bills	Dashboard, financial reports, CSV, private bill attachments, permission checks on exports	Treasurer can produce and verify a festival-wise report
M6: Pilot (Weeks 7–8)	Deploy and test in village	HTTPS deployment, backups and restore test, training of 1–2 members, parallel run with notebook, feedback log	Temple can use the app without developer help for each transaction

15. Pilot Plan and Acceptance Criteria

Pilot approach: run the app *in parallel* with the paper notebook for at least 2 to 4 weeks. Compare totals weekly. Retire the notebook only when the committee agrees.

Acceptance criteria (all must pass)

- A committee member records a donation and an expense without developer help.
- Dashboard totals match the notebook and actual cash/bank for the pilot period.
- A festival-wise report is generated and checked by the treasurer.
- A villager finds a correct festival date and announcement on their phone.
- Temple A data cannot be read or changed by Temple B users (tested with two test temples).
- Anonymous visitors only see published information.
- Editing and reversing a record preserves the audit trail.

- Duplicate submissions do not create duplicate entries.
- Backup and restore have been tested successfully.
- Committee members confirm they prefer the app after the trial.

Release gate checklist (before real money data is entered)

1. Tenant isolation tested.
2. Role permissions tested (treasurer vs admin).
3. Server-side validation of amounts, dates, uploads.
4. HTTPS enabled; passwords hashed; sessions secure.
5. Donor data and bills not exposed publicly or in logs.
6. Backups running and restore proven.
7. Opening balance documented and signed off by the committee.

16. Success Metrics

Metric	Pilot target
Share of new donations/expenses recorded in the app	100% after the parallel-run period
Reconciliation difference vs. real cash/bank	Zero unexplained difference
Time to record a donation	Under 30 seconds
Committee members able to operate independently	At least 2
Critical data-access bugs	0
Villager satisfaction with festival information	Positive feedback from a sample of villagers

17. Risks and Mitigations

Risk	Impact	Mitigation
Committee does not trust or adopt the app	High	Involve them from week 1; parallel run; simple UI; training

Risk	Impact	Mitigation
Incorrect festival dates published	High	Use verified sources only; pandit review before publishing; show source and verification status
Data leak between temples	Critical	Server-side tenant checks; automated isolation tests; release gate
Financial data loss	Critical	Daily backups, tested restore, no hard deletes
Data entry mistakes	Medium	Validation, confirmation step, reversals with audit
Poor internet in village	Medium	Lightweight pages; consider draft-saving/offline in 1.x
Low digital literacy	Medium	Hindi UI, large buttons, in-person training, short how-to guide
Misuse by an insider	Medium	Role limits, audit log, committee-level review of reports
Developer dependency	Medium	Documentation, OpenAPI, clean modular code, runbook
Scope creep (AI, payments too early)	Medium	Strict out-of-scope list for V1; roadmap-based phases
Legal/trust matters (donation records, privacy norms)	Medium	Confirm local requirements with the committee; minimal donor data; privacy note

18. Roadmap After V1

Release	Theme	Key items
1.1	Donations	Display official UPI QR; manual verification; later, payment provider with verified confirmations and reconciliation

Release	Theme	Key items
1.2	Multi-temple	Controlled self-registration, temple verification, member invitations, platform admin tools
1.3	Engagement	SMS/WhatsApp alerts, offline drafts, more roles (auditor)
2.0	AI assistant	Text assistant on verified data, then voice in Hindi and Bhojpuri with evaluation of accuracy before wide release
Later	Scale	More languages, other regional calendars, optional Android app

19. Open Questions and Decisions Needed

1. Final product name and domain availability (MandirSetu or other).
2. Which login method suits committee members best (email, phone number, or both)?
3. Which regional calendar and tradition should the first festival data follow, and who verifies it?
4. Who in the committee is allowed to handle money, and what approvals do they want?
5. Should donor names be public, private, or per-donor choice at this temple?
6. What is the exact opening balance (cash and bank) and its start date?
7. Hosting provider and budget for the pilot (hosting, domain, backup storage).
8. Will receipts be printed or shared digitally (for example, as image/PDF)?
9. Who will provide first-line support to committee members after launch?
10. Who owns the product legally, and what happens to the data if the developer changes?

20. Next Three Actions

1. **Meet the committee.** Document how they record chanda today, how they total balances and who handles the money. Show them Section 1 of this document and get their feedback and agreement.
2. **Collect pilot data.** Temple profile, expense categories, opening balance details, verified festival list for the year.

3. **Start M1 and M2.** Repository, database, login, roles and tenant isolation first; donations and expenses next.

This PRD defines the proposed Version 1 contracts and scope. API formats, screens and schedules will be refined with the development team and the temple committee during the build.